Below is a **clear, step-by-step guide to file handling with PDF files in Python**, explained **from basics to practical usage**, with **simple concepts**, **working examples**, and **real-world tips**.

Important idea first:
PDF is NOT a plain text file.
You cannot reliably use open("file.pdf", "r") like .txt.

1️⃣ How PDF Handling Works in Python (Easy Concept)

PDF files are **binary + structured documents**.
So Python uses **special libraries** to:

- Read text from PDFs
- Extract pages
- Create new PDFs
- Merge or split PDFs

2️⃣ Most Common PDF Libraries

Library	Purpose	Level
PyPDF2	Read, merge, split PDFs	Beginner
pdfplumber	Accurate text extraction	Intermediate
reportlab	Create PDFs	Advanced
fpdf	Simple PDF creation	Beginner

👉 We'll focus on **PyPDF2** (most common).

3️⃣ Install Required Library

```
pip install PyPDF2
```

Verify:

```
import PyPDF2
```

```
print(PyPDF2.__version__)
```

4️⃣ Reading a PDF File (Basic)

```
import PyPDF2

file = open("sample.pdf", "rb")   # rb = read binary
reader = PyPDF2.PdfReader(file)

print(len(reader.pages))  # total pages
```

📌 PDFs must be opened in **binary mode (rb)**.

5️⃣ Read Text from a PDF Page

```
page = reader.pages[0]
text = page.extract_text()
print(text)

file.close()
```

✓ Reads visible text
❌ Images/text-as-images won't be extracted

6️⃣ Read All Pages of a PDF

```
with open("sample.pdf", "rb") as file:
  reader = PyPDF2.PdfReader(file)

  for page in reader.pages:
    print(page.extract_text())
```

✓ Best practice
✓ Automatically closes file

7️⃣ Write / Create a New PDF (Basic)

```
from PyPDF2 import PdfWriter

writer = PdfWriter()

with open("sample.pdf", "rb") as file:
    reader = PyPDF2.PdfReader(file)
    writer.add_page(reader.pages[0])

with open("output.pdf", "wb") as output:
    writer.write(output)
```

✓ Creates new PDF
✓ Writes selected pages

8️⃣ Merge Multiple PDFs

```
from PyPDF2 import PdfMerger

merger = PdfMerger()
merger.append("file1.pdf")
merger.append("file2.pdf")

merger.write("merged.pdf")
merger.close()
```

9️⃣ Split PDF Pages

```
from PyPDF2 import PdfWriter, PdfReader

reader = PdfReader("sample.pdf")

writer = PdfWriter()
writer.add_page(reader.pages[0])

with open("page1.pdf", "wb") as output:
  writer.write(output)
```

🔟 Extract PDF Text and Save to .txt

```
with open("sample.pdf", "rb") as pdf:
  reader = PyPDF2.PdfReader(pdf)

  with open("output.txt", "w", encoding="utf-8") as text_file:
    for page in reader.pages:
      text_file.write(page.extract_text())
```

✓ Very common real-world task

1️⃣1️⃣ Create PDF from Scratch (Simple)

Using fpdf

```
pip install fpdf
from fpdf import FPDF
```

```
pdf = FPDF()

pdf.add_page()

pdf.set_font("Arial", size=12)

pdf.cell(200, 10, txt="Hello PDF", ln=True)


pdf.output("new.pdf")
```

1️⃣2️⃣ Common Mistakes (IMPORTANT)

Mistake	Problem
Opening PDF with "r"	❌ Must use "rb"
Expecting perfect text	PDFs are layout-based
Reading scanned PDFs	Needs OCR (Tesseract)
Forgetting to close file	Memory leak

1️⃣3️⃣ When PDF Text Is NOT Readable

PDF might be **image-based** (scanned).

Solution:

- Use **OCR**
- Libraries:
  - pytesseract
  - pdf2image

1️⃣4️⃣ Real-World Use Cases

✓ Resume parser
✓ Invoice extraction

✓ Report automation
✓ PDF → TXT conversion
✓ PDF archiving systems

1️⃣ 5️⃣ Mini Practice Tasks (Do These)

1. Read a PDF and print page count
2. Extract first page text
3. Merge two PDFs
4. Save PDF text to .txt
5. Create a simple PDF file

📌 Final Rule (Remember Forever)

TXT → open()

PDF → library

If you want next:

- 🔍 OCR scanned PDFs
- 📊 PDF → Excel conversion
- 🧠 PDF interview questions
- 🔄 Compare PDF vs TXT vs Excel
- 🛠️ Build a PDF automation project

Just tell me 👍